

ON NEUROMORPHIC COMPUTING

WITH SPIKING NEURAL NETWORKS

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INTRODUCTION AND MOTIVATION

- **The von Neumann Bottleneck:** The physical separation of processing and memory forces massive energy expenditure on data transport[cite: 1].
- Fetching weights from DRAM consumes orders of magnitude more energy than the computation itself[cite: 1].
- **Dense Processing of Sparse Data:** Deep learning hardware executes dense matrix multiplications regardless of activation sparsity, wasting cycles on null results[cite: 1].
- **The Cost of Synchrony:** Global clock networks consume up to 40% of power budgets and enforce worst-case latencies[cite: 1].
- **Biological Contrast:** The brain operates on 20W, performing real-time multi-sensory reasoning through asynchronous, sparse events[cite: 1].

lets draw more inspiration from biological intelligence → neuromorphic computing

WHY IS NEUROMORPHIC MORE EFFECIENT?

SHIFTING THE COMPUTING PARADIGM

Neuromorphic engineering replaces rigid, clock-driven logic with the adaptive, event-driven dynamics of neural tissue[cite: 1].

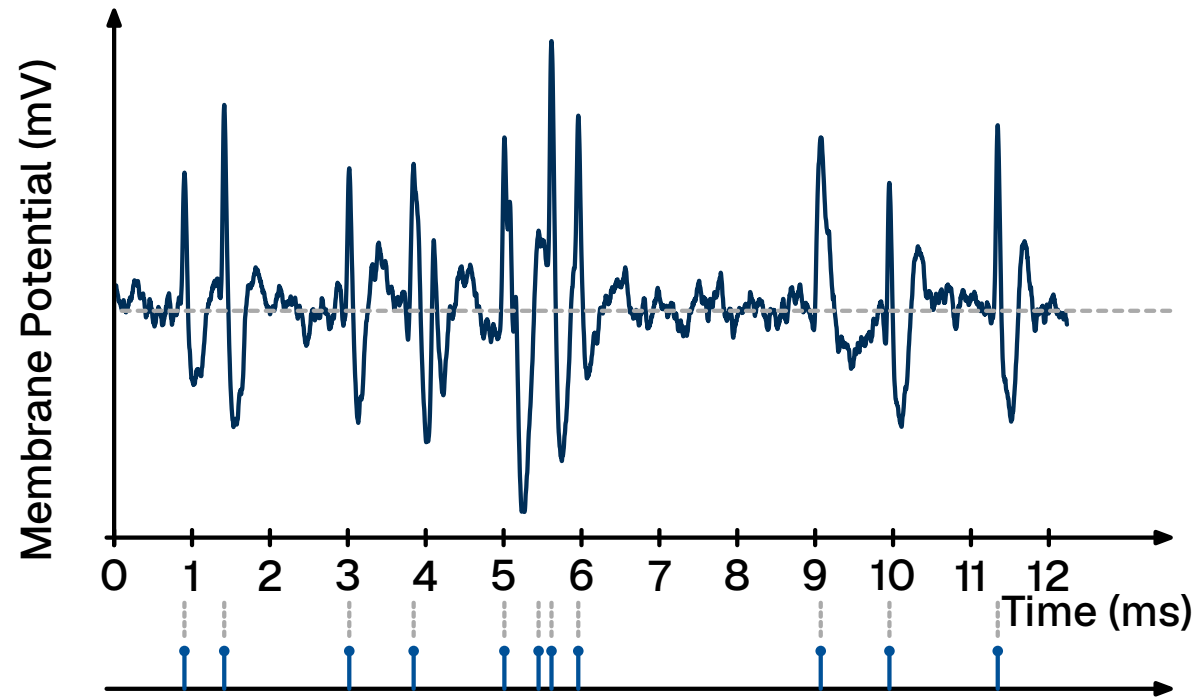
- **The Synaptic Principle:** Eliminates the von Neumann bottleneck by physically co-locating memory and compute across the silicon die[cite: 1].
- **The Action Potential Principle:** Operates asynchronously, driven strictly by the arrival of data, ensuring energy scales linearly with task complexity[cite: 1].
- **The Spike Principle:** Communicates via discrete binary events, drastically reducing bandwidth by encoding information in precise timing rather than complex magnitudes[cite: 1].

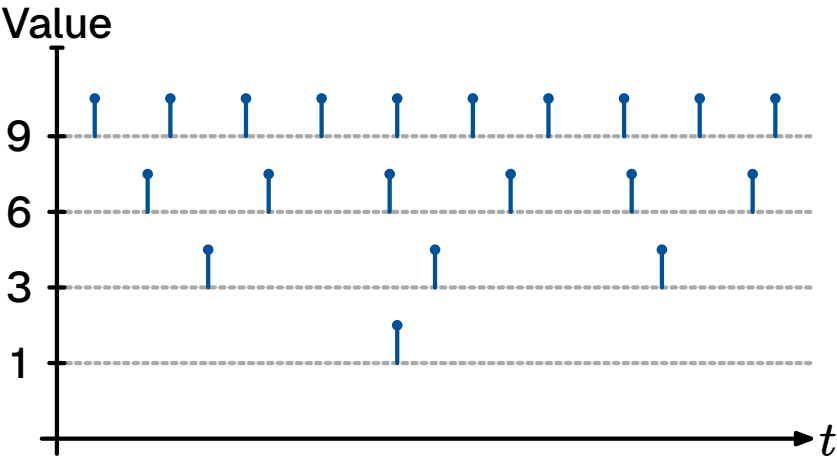
Research Objectives

Core Questions Addressed in this Thesis

- **Sparse Efficient Computing:** Do biologically inspired, event-driven algorithms reduce the computational footprint of visual classification?[cite: 1]
- **Neuron Model Evaluation:** Which temporal integration dynamics are compatible with Time-to-First-Spike (TTFS) decoding?[cite: 1]
- **Inference Via Weight Transfer:** What is the accuracy penalty of zero-shot ANN-to-SNN weight transfer under TTFS?[cite: 1]
- **Native Unsupervised Learning:** Can local Spike-Timing-Dependent Plasticity (STDP) autonomously extract meaningful geometric features?[cite: 1]

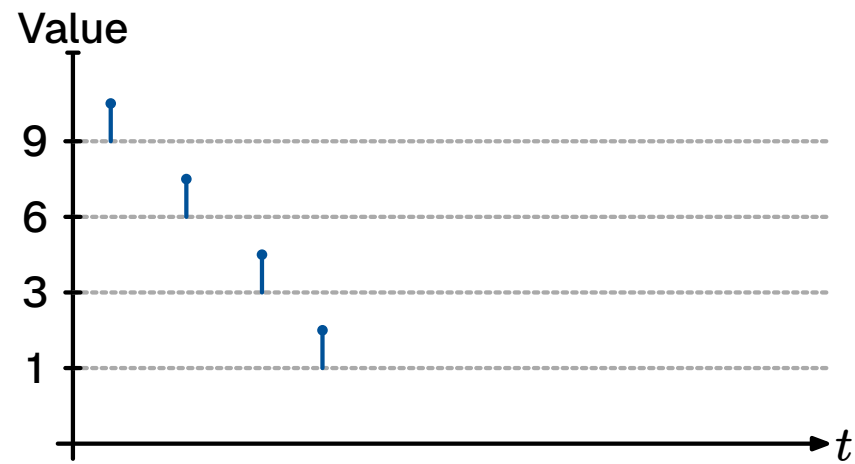
ENCODING INFORMATION IN SPIKES





Time-to-First-Spike (TTFS) Encoding

- Unlike Rate Coding, which requires extended time windows, TTFS encodes stimulus intensity inversely to response latency[cite: 1].
- **Mechanism:** A high-intensity (bright) pixel triggers an early spike, compressing spatial information into a priority-driven queue[cite: 1].
- **Advantages:**
 - **Sparsity:** Sub-threshold noise (background pixels) is aggressively discarded and never fires[cite: 1].
 - **Latency:** Eliminates the need to wait for a time window to close; processing begins as soon as salient features arrive[cite: 1].
- Phase ambiguity is resolved by simulating a biological saccade, establishing a global temporal reference frame (t_0)[cite: 1].



Phase I: Neuron Model Dynamics

Decoding Temporal Sequences

- Evaluated four architectures (IF, LIF, Linear Ramp, State Discount) across saturation, critical, and deficit threshold regimes[cite: 1].
- **The LIF Misalignment:** Under critical constraints, the standard Leaky Integrate-and-Fire model actively penalized early, high-salience spikes due to its exponential decay[cite: 1].
- **The Solution (Model C):** Developed a Current-Accumulating Linear Ramp model combining integration momentum with a strict 10-tick coincidence window[cite: 1].
- Model C successfully preserved rank-order sequence priority without the computational overhead of continuous exponentials[cite: 1].

Methodology & Setup

Experimental Framework

- **Dataset:** MNIST (28x28 normalized grayscale images), chosen for its high degree of spatial sparsity[cite: 1].
- **Topology:** Fully Connected Network (FCN) $784 \rightarrow 128 \rightarrow 10$ [cite: 1].
 - An FCN allows mathematically transparent, one-to-one parameter mapping without the structural overhead of convolutional unrolling[cite: 1].
- **SNN Simulator:** Custom PyTorch discrete-time engine operating over a $T_{\{\max\}} = 64$ tick saccade window[cite: 1].
- **Metrics:** Top-1 Accuracy, Temporal Latency (Time-to-Decision), and Synaptic Operations (SyOPs) as a hardware proxy[cite: 1].

Phase II: Zero-Shot Weight Transfer

Translating ANN to SNN

- A baseline ANN was trained without bias terms to 98.40% accuracy, establishing an ideal ceiling[cite: 1].
- Continuous FP32 weights were transferred **directly** to the SNN (Model C) without any intermediate retraining or fine-tuning[cite: 1].
- **Results:**
 - **Accuracy:** Maintained 94.50%, proving momentum-based TTFS successfully preserves spatial hierarchies[cite: 1].
 - **Latency:** Reached decisions rapidly, with a mean latency of 8.4 ticks[cite: 1].
 - **Efficiency:** Achieved an 85.2% reduction in operations (SyOPs) due to the early-exit S-curve profile[cite: 1].

Phase III: Native Unsupervised Learning

Local STDP & Self-Organization

- Discarded pre-trained weights; initialized randomly and trained strictly via a discrete, vectorized STDP rule[cite: 1].
- **Competitive Specialization:**
 - **Winner-Takes-All (WTA):** The first output neuron to fire suppresses all competitors, forcing distinct geometric clustering[cite: 1].
 - **Homeostasis:** Implemented adaptive threshold decay and winner penalties to prevent single-neuron dominance (dead neurons)[cite: 1].

Phase III: Results & Limitations

Representational Boundaries

- **Accuracy:** Achieved 44.3% top-1 accuracy via unsupervised self-organization[cite: 1].
- Successfully clustered geometric features without global error signals, maintaining the early-exit sparsity (87.2% compute reduction)[cite: 1].
- **The Limitation:** Single-layer WTA forces rigid, holistic template matching[cite: 1].
- The network struggled with morphological overlaps (e.g., '5' vs. '0') because the global WTA discards complementary spatial evidence in favor of the single most salient feature[cite: 1].

Discussion

The Sparsity Paradox

While theoretical SynOps were reduced by $> 85\%$, simulating this on von Neumann hardware highlights the central engineering bottleneck.

- Deep learning frameworks and GPUs are optimized for dense matrix multiplications and will execute floating-point cycles even for boolean-masked zeros[cite: 1].
- Combined with temporal loop overhead, the SNN simulator inherently consumes more absolute clock cycles on a GPU than the continuous ANN baseline[cite: 1].
- **Conclusion:** Realizing calculated energy gains requires deployment on native event-driven substrates (e.g., neuromorphic ASICs or FPGAs) that utilize Address Event Representation (AER) to physically halt processing[cite: 1].

Future Work

Bridging the Engineering Gap

- **Native Sensors:** Transitioning to Dynamic Vision Sensors (DVS) to encode contrast changes directly, bypassing the latency bottlenecks of processing dark pixels under TTFS[cite: 1].
- **Spiking CNNs:** Replacing global WTA with localized receptive fields and local lateral inhibition to build deep, translation-invariant hierarchies[cite: 1].
- **Structural Plasticity:** Pruning depressed synapses to create block-sparse tensors, yielding physical memory bandwidth reductions even prior to ASIC deployment[cite: 1].

Conclusion

- The standard deep learning scaling paradigm is approaching strict physical limits[cite: 1].
- This work demonstrates that TTFS temporal encoding drastically reduces computational workloads via early-exit, momentum-based integration[cite: 1].
- While local STDP is viable for unsupervised feature extraction, unlocking the true potential of these algorithms requires bridging the gap between event-driven software and specialized physical hardware[cite: 1].